

Answer the following (short answers)

- a) Compare holozoic, saprophytic and parasitic nutrition. Give one example of each
- b) Why do holozoic organisms need a digestive system but saprophytic don't?
- c) If we remove all heterotrophs from the earth, will autotrophs survive? Why/why not?
- d) Pepsin works in stomach but not in small intestine. Trypsin works in small intestine but not in stomach. Why?
- e) A man eats only boiled potatoes for one month. Which digestive enzymes will be used most? Which will be least used?

Answer the following (long answers)

- a) Draw and label the parts of stomata and explain how it opens and closes.
- b) How do unicellular organisms obtain their nutrition? Explain the process diagrammatically in amoeba and paramecium.
- c) In an experiment to show that CO₂ is necessary for photosynthesis
 - i) Why is the plant kept in dark before experiment?
 - ii) Why do we boil leaf in water?
 - iii) Why do we boil leaf in alcohol in water bath, not directly on flame?
 - iv) Which gas is removed by KOH?
 - v) Why is vaseline applied on bell jar?
- d) Human digestive system is a tube running from mouth to anus. Its main function is to breakdown complex molecules present in the food which cannot be absorbed as such into smaller molecules. These molecules are absorbed across the walls of the tube and the absorbed food reaches each and every cell of body where it is utilized for obtaining energy.
 - i) Name the glands present in the buccal cavity and write the components of food on which the secretion of these glands act upon.
 - ii) Two organs have a sphincter muscles at their exit. Name them.

iii) What will happen if mucus is not secreted by the gastric glands?

iv) What will happen if villi are absent in the small intestine?

OR

ii) *Bile juice does not contain any enzyme. Yet, it has important roles in digestion." Justify the statement and name the organ that produces the bile.

e) Make 50 MCQs from the chapter 5